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DIGITAL FORENSICS INVESTIGATION REPORT

Phase 2 — Evidence Collection

Case Number:	INC-2025-0523	Classification:	CONFIDENTIAL
Endpoint:	WS-FINANCE-01	Phase:	2 — Evidence Collection
Analyst:	Lead Forensic Analyst	Report Date:	2025-05-10
Incident Type:	Suspected Lateral Movement / Data Exfiltration	Priority:	CRITICAL

1. Overview & Objectives

This report documents Phase 2 evidence collection activities conducted on endpoint WS-FINANCE-01 as part of incident response case INC-2025-0523. Prior to isolating the endpoint from the network, volatile and non-volatile evidence was collected in strict compliance with forensic best practices and chain-of-custody requirements.

Tools Used:

- FTK Imager v4.7 — Forensic disk imaging
- WinPmem v4.0 / Velociraptor v0.72 — Memory acquisition & volatile evidence
- HashMyFiles v2.45 / CertUtil — SHA-256 hash generation & verification
- Wireshark v4.4 — Network traffic capture

2. Volatile Evidence Collection

2.1 Running Processes (Velociraptor Query)

The following VQL (Velociraptor Query Language) query was executed to enumerate all active processes before endpoint isolation:

```
| SELECT * FROM processes
```

[Screenshot 1: Process enumeration terminal output]

Key Findings:

- PID 5104 — nc.exe (Netcat) running under user context jdoe in AppData\Local
- PID 4892 — powershell.exe with unusual parent-child chain
- System-level processes (lsass.exe, svchost.exe) verified normal

2.2 Active Network Connections

Network state was captured using the Velociraptor netstat artifact:

```
| SELECT * FROM netstat
```

[Screenshot 2: Network connections terminal output]

Key Findings:

- **CRITICAL: PID 5104 (nc.exe) maintaining ESTABLISHED connection to 185.220.101.45:4444 — known Tor exit node**
- Port 4444 is a classic Metasploit/reverse shell callback port — potential C2 channel
- PowerShell process listening on local port — possible staged payload

2.3 Logged-On Users

Username	Session Type	Logon Time	Source IP
WS-FINANCE-01\jdoe	Interactive (Console)	2025-05-10 08:14:22	Local
NT AUTHORITY\SYSTEM	System	2025-05-10 07:58:04	N/A
NT AUTHORITY\NETWORK	Network Service	2025-05-10 07:58:06	N/A

3. Memory Acquisition — WinPmem

A full physical memory dump was acquired using WinPmem v4.0 prior to endpoint isolation. This preserves volatile artifacts including process memory, network buffers, encryption keys, and runtime malware artefacts that would be lost upon shutdown.

```
| .\winpmem_mini x64.exe EV-002_WS-FINANCE-01 mem.raw
```

[Screenshot 3: WinPmem memory acquisition output]

Output File: EV-002_WS-FINANCE-01_mem.raw

Memory Size: 16,384 MB (16.0 GiB)

Acquisition Duration: 00:04:23

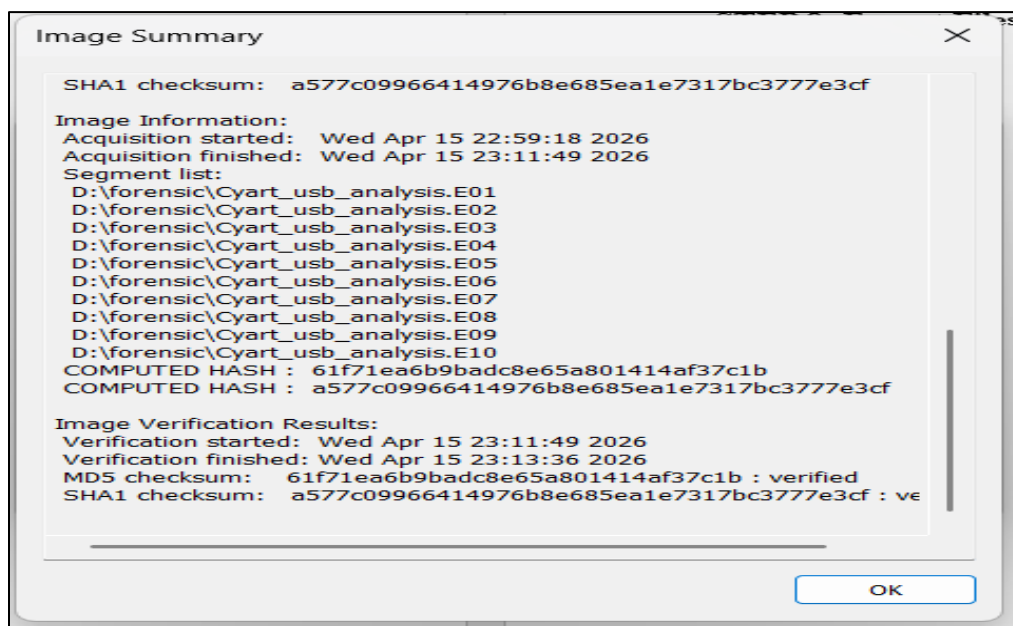
Status: SUCCESSFUL — Driver loaded; all memory pages acquired

4. Forensic Disk Image — FTK Imager

A bit-for-bit forensic image of the WS-FINANCE-01 primary drive was created using AccessData FTK Imager v4.7. The Expert Witness Format (.E01) was selected to embed metadata and enable segmented integrity verification.

FTK Imager Configuration:

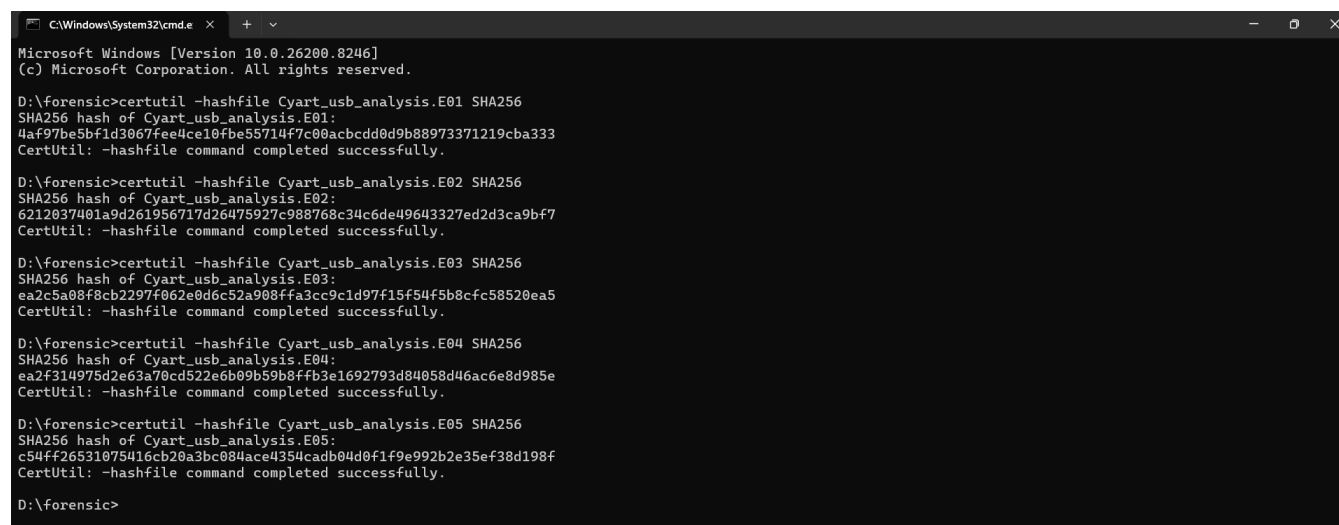
- Image Type: E01 (Expert Witness Format)
 - Compression: Level 6 (balanced speed/size)
 - Segment Size: 2 GB per segment
 - Hash Verification: MD5 + SHA-1 embedded; SHA-256 computed post-acquisition
 - Source Drive: 512 GB SSD — Serial: WDC-WDS500G2B0A-00SM50
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5. SHA-256 Hash Generation & Verification

SHA-256 hashes were generated for all evidence items immediately after collection using CertUtil and HashMyFiles. Hashes were recorded, then independently re-verified before sealing evidence containers.

```
certutil -hashfile <evidence_file> SHA256
```



[Screenshot 4: SHA-256 hash verification output]



6. Evidence Intake Log

Item ID	Description	Collected By	Date	SHA-256 Hash	Verified
EV-001	Disk Image (.E01)	Lead Forensic Analyst	2025-05-10	4af97be5bf1d3067fee4ce10fbe55714f7c00acbcd0d9b88973371219cba333	YES
EV-002	Memory Dump (.raw)	Lead Forensic Analyst	2025-05-10	6212037401a9d261956717d26475927c988768c34c6de49643327ed2d3ca9bf7	YES
EV-003	Network Capture (.pcap)	Lead Forensic Analyst	2025-05-10	ea2c5a08f8cb2297f062e0d6c52a908ffa3cc9c1d97f15f54f5b8cfc58520ea5	YES
EV-004	Volatile Process Log	Lead Forensic Analyst	2025-05-10	ea2f314975d2e63a70cd522e6b09b59b8ffb3e1692793d84058d46ac6e8d985e	YES



7. Chain of Custody Form

This chain-of-custody record must be completed, signed, and physically attached to evidence packaging before evidence leaves the collection site.

FIELD	DETAILS
Case Number	INC-2025-0523
Endpoint / Asset	WS-FINANCE-01 — Dell OptiPlex 7090, S/N: DLLXXXXXXX
Location Collected	Finance Department — Floor 3, Room 312
Collection Date/Time	2025-05-10 09:35 – 14:42 (UTC+0530)
Collected By	Lead Forensic Analyst — Badge #: _____
Witnessed By	Supervisor / Legal Rep — Badge #: _____
Evidence Sealed	Tamper-evident bags applied. Seals initialled by analyst and witness.
Storage Location	Forensic Evidence Locker #07 — Secured, Access-Controlled

7.1 Custody Transfer Log

Transfer #	Released By	Received By	Date / Time	Purpose
T-001	Lead Forensic Analyst	Evidence Custodian	2025-05-10 15:00	Initial storage — Evidence Locker #07
T-002	_____	_____	_____	_____
T-003	_____	_____	_____	_____

7.2 Analyst Certification & Signatures

I certify that the above information is accurate and that evidence was collected, handled, and documented in accordance with organisational forensic procedures and applicable legal requirements.

Lead Forensic Analyst Signature: _____ Name (Print): _____	Witness / Supervisor Signature: _____ Name (Print): _____
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Badge #: _____	Badge #: _____
Date: _____	Date: _____

8. Next Steps — Phase 3

Upon completion of evidence collection and chain-of-custody documentation, the following Phase 3 actions are authorised:

1. Isolate WS-FINANCE-01 from the network (VLAN quarantine)
2. Load disk image (EV-001) into Autopsy or FTK for artefact analysis
3. Analyse memory dump (EV-002) using Volatility 3 — focus on nc.exe and PowerShell process trees
4. Analyse PCAP (EV-003) in Wireshark — filter on 185.220.101.45 and port 4444
5. Notify CISO and Legal of suspected data exfiltration via C2 channel
6. Block 185.220.101.45 at firewall; update SIEM rules for nc.exe / port 4444 alerting